ASCE 41-23 Tier 1 Checklists

FIRM:	
PROJECT NAME:	
SEISMICITY LEVEL:	
PROJECT NUMBER:	
COMPLETED BY:	
DATE COMPLETED:	
REVIEWED BY:	
REVIEW DATE:	

Legend: C = Compliant, NC = Noncompliant, N/A = Not Applicable, U = Unknown

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Project Name _____
Project Number _____

17.6IO Structural Checklist for Building Type S3: Metal Building Frames

Table 17-13. Immediate Occupancy Checklist for Building Type S3.

Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
Very Low and Low Seismicity							
Seismic-Force-Resisting System							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	BRACE AXIAL STRESS CHECK: The axial stress in the diagonals, calculated using the Quick Check procedure of Section 4.4.3.4 is less than $0.50F_y$.	5.5.4.1	A.3.3.1.2	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	FLEXURAL STRESS CHECK: The average flexural stress in the moment-frame columns and beams, calculated using the Quick Check procedure of Section 4.4.3.9 is less than F_y.	5.5.2.2.2	A.3.1.3.3	
Connections							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	TRANSFER TO STEEL FRAMES: Diaphragms are connected for transfer of seismic forces to the steel moment frames.	5.7.2	A.5.2.2	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	STEEL COLUMNS: The columns in seismic-force-resisting frames are anchored to the building foundation with a minimum of two anchor rods and with the base plates bearing on concrete or a grout pad. The anchor rods are capable of resisting the overturning force using the Quick Check procedure of Section 4.4.3.6.	5.7.3.1	A.5.3.1	

Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
Moderate Seismicity (Complete the Following Items in Addition to the Items for Very Low and Low Seismicity)							
Seismic-Force-Resisting System							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	MOMENT-RESISTING CONNECTIONS: All moment connections are able to develop the elastic moment (F_yS) of the adjoining members.	5.5.2.2.1	A.3.1.3.4	
Diaphragms							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	PLAN IRREGULARITIES: There is tensile capacity to develop the strength of the diaphragm at reentrant corners or other locations of plan irregularities.	5.6.1.4	A.4.1.7	

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Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
C	NC	N/A	U	DIAPHRAGM REINFORCEMENT AT OPENINGS: There is reinforcing around all diaphragm openings larger than 50% of the building width in either major plan dimension.	5.6.1.5	A.4.1.8	
C	NC	N/A	U	OTHER DIAPHRAGMS: Diaphragms do not consist of a system other than wood, steel deck, concrete, or horizontal bracing.	5.6.5	A.4.7.1	
Connections							
C	NC	N/A	U	ROOF PANELS: Where considered as diaphragm elements for lateral resistance, metal, plastic, or cementitious roof panels are positively attached to the roof framing to resist seismic forces.	5.7.5	A.5.5.1	
C	NC	N/A	U	WALL PANELS: Where considered as shear elements for lateral resistance, metal, fiberglass, or cementitious wall panels are positively attached to the framing and foundation to resist seismic forces.	5.7.5	A.5.5.2	
High Seismicity (Complete the Following Items in Addition to the Items for Low and Moderate Seismicity)							
Seismic-Force-Resisting System							
C	NC	N/A	U	MOMENT-RESISTING CONNECTIONS: All moment connections are able to develop the strength of the adjoining members or panel zones.	5.5.2.2.1	A.3.1.3.4	
C	NC	N/A	U	COMPACT MEMBERS: All frame elements meet compact section requirements in accordance with AISC 360, Table B4.1a.	5.5.2.2.5	A.3.1.3.8	
C	NC	N/A	U	BEAM PENETRATIONS: All openings in frame-beam webs are less than one-quarter of the beam depth and are located in the center half of the beams.	5.5.2.2.6	A.3.1.3.9	
C	NC	N/A	U	OUT-OF-PLANE BRACING: Beam–column joints are braced out of plane.	5.5.2.2.8	A.3.1.3.11	
C	NC	N/A	U	BOTTOM FLANGE BRACING: The bottom flanges of beams are braced out of plane.	5.5.2.2.9	A.3.1.3.12	

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Connections						
C	**NC**	**N/A**	**U**	TRANSFER TO STEEL FRAMES: Diaphragms are connected for transfer of seismic forces to the steel moment frames, and the connections are able to develop the lesser of the strength of the frames or the diaphragms.	5.7.2	A.5.2.2
☐	☐	☐	☐			
C	**NC**	**N/A**	**U**	STEEL COLUMNS: The columns in seismic-force-resisting frames are anchored to the building foundation, and the anchorage is able to develop the least of the following: the tensile capacity of the column, the tensile capacity of the lowest-level column splice (if any), or the uplift capacity of the foundation.	5.7.3.1	A.5.3.1
☐	☐	☐	☐			

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